QuAIR Research Notes **2026**

Note 14: Generating Clifford Group

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14.1 Summary of Progress

1. Analyzed single-qubit Clifford operators and verified $\overline{\mathcal{C}}_1$ is generated by H and S.
2. Built the symplectic representation of $\mathcal{C}_n$ and designed the SBGR algorithm for $\mathrm{Sp}(2n, \mathbb{F}_2)$.
3. Derived the short exact sequence and proved multi-qubit projective Clifford group is generated by $H_j, S_j, \mathrm{CNOT}_{i\to j}$.

14.2 Clifford Group

Recall Notation:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix};$$

$$\mathcal{P}_n := \left\{ i^k X^{a_1} Z^{b_1} \otimes \cdots \otimes X^{a_n} Z^{b_n} \mid k \in \mathbb{Z};\ a_j, b_j \in \{0,1\} \right\}$$

Recall the fundamental Pauli commutation relations

$$XY = iZ, \quad YZ = iX, \quad ZX = iY.$$

Definition 1 Define the **full Clifford group** as the normalizer of Pauli group

$$\mathcal{C}_n := N(\mathcal{P}_n) := \left\{ U \in \mathrm{U}(2^n) \mid \forall\, P \in \mathcal{P}_n,\, UPU^\dagger \in \mathcal{P}_n \right\};$$

Define the **projective Clifford group** via quotienting out global phases:

$$\overline{\mathcal{C}}_n := N(\mathcal{P}_n) / \langle e^{i\phi} I : \phi \in \mathbb{R} \rangle = N(\mathcal{P}_n)/\mathrm{U}(1);$$

Define the **projective Pauli group** correspondingly:

$$\overline{\mathcal{P}}_n := \{\lambda P \mid \lambda \in \mathrm{U}(1),\, P \in \mathcal{P}_n\} / \mathrm{U}(1) = \left\{ \overline{P} \mid P \in \mathcal{P}_n \right\};$$

Define the **abstract Clifford group** as the second quotient:

$$\overline{\mathcal{C}}_n^{\mathrm{abs}} := \overline{\mathcal{C}}_n / \overline{\mathcal{P}}_n.$$

Remark 2 Obviously $e^{i\phi} I \in N(\mathcal{P}_n)$ and $\mathrm{U}(1)\mathcal{P}_n \leq \mathcal{C}_n$. Thus

$$\begin{aligned} \overline{\mathcal{C}}_n^{\mathrm{abs}} &= \overline{\mathcal{C}}_n / \overline{\mathcal{P}}_n \\ &= \frac{\mathcal{C}_n/\mathrm{U}(1)}{\{\lambda P \mid P \in \mathcal{P}_n,\, \lambda \in \mathrm{U}(1)\}/\mathrm{U}(1)} \\ &\simeq \mathcal{C}_n / \{\lambda P \mid P \in \mathcal{P}_n,\, \lambda \in \mathrm{U}(1)\} \\ &= \mathcal{C}_n / \big(\mathrm{U}(1)\mathcal{P}_n\big). \end{aligned}$$

Lemma 3 *Let $\{X, Y, Z\} = \{P_1, P_2, P_3\}$. Suppose $U \in \mathrm{U}(2)$ such that $UPU^\dagger \in \mathcal{P}_1$. Then*

1. *$UP_iU^\dagger = \pm P_j$ for some index j.*
2. *If $UP_iU^\dagger = \pm P_i$, then*
   (i) $UP_iU^\dagger = P_i \implies U = \lambda I$ or $U = \lambda P_i$;
   (ii) $UP_iU^\dagger = -P_i \implies U = aP_k + bP_\ell,\ k, \ell \neq i$.
3. *$UP_iU^\dagger = \lambda_i P_{\sigma(i)}$, where the induced map $\sigma \in S_3$ is a permutation on $\{1, 2, 3\}$.*

Proof **(1)** Since $P_i^2 = P_j^2 = I$. Suppose $UP_iU^\dagger = cP_j$ for some index j and constant $c \in \mathbb{C}$. Square both sides:

$$\left(UP_iU^\dagger\right)^2 = c^2 P_j^2.$$

$$\left(UP_iU^\dagger\right)^2 = UP_iU^\dagger UP_iU^\dagger = UP_i^2U^\dagger = UIU^\dagger = I = c^2P_j^2 = c^2 I.$$

So $c^2 = 1$, which gives $c = \pm 1$. Hence

$$UP_iU^\dagger = \pm P_j.$$

(2) Every 2×2 complex matrix has unique Pauli basis expansion:

$$U = aI + bP_1 + cP_2 + dP_3, \quad a, b, c, d \in \mathbb{C}.$$

(i) Commutation case $UP_3 = P_3U$ (take $i = 3$ without loss of generality)

$$\begin{aligned} UP_3 - P_3U &= \big(aI + bP_1 + cP_2 + dP_3\big)P_3 - P_3\big(aI + bP_1 + cP_2 + dP_3\big) \\ &= b(P_1P_3 - P_3P_1) + c(P_2P_3 - P_3P_2) \\ &= b(-2iP_2) + c(2iP_1). \end{aligned}$$

Set equal to zero: $2icP_1 - 2ibP_2 = 0 \implies b = 0,\ c = 0$.

Thus $U = aI + dP_3$. Since U is Unitary,

$$I = UU^\dagger = (aI + dP_3)(\overline{a}I + \overline{d}P_3) = (|a|^2 + |d|^2)I + a\overline{d}P_3 + \overline{a}dP_3.$$

So cross term coefficient vanishes, either $a = 0$ or $d = 0$, which reduces it to $U = \lambda I$ or $U = \lambda P_3$ with $|\lambda| = 1$.

(ii) Anti-commutation case $UP_3 = -P_3U$ (take $i = 3$ without loss of generality)

$$\begin{aligned} UP_3 + P_3U &= \big(aI + bP_1 + cP_2 + dP_3\big)P_3 + P_3\big(aI + bP_1 + cP_2 + dP_3\big) \\ &= 2aP_3 + b(P_1P_3 + P_3P_1) + c(P_2P_3 + P_3P_2) + 2dP_3^2 \\ &= 2aP_3 + 2dI. \end{aligned}$$

Equate to zero: $2dI + 2aP_3 = 0 \implies a = 0,\ d = 0$. Only cross terms remain:

$$U = bP_1 + cP_2.$$

For general index i, we have $U = aP_k + bP_\ell$ for distinct indices $k, \ell \neq i$.

(3) Define mapping $\phi(P_i) = UP_iU^\dagger$. If $UP_iU^\dagger = \lambda UP_{i'}U^\dagger$ for some $\lambda \in \mathbb{C}$, multiply both sides by $U^\dagger$ on the left and U on the right:

$$P_i = \lambda P_{i'}.$$

Since P_1, P_2, P_3 are linearly independent over $\mathbb{C}$, we must have $i = i'$. Thus there exists a permutation $\sigma \in S_3$ such that

$$UP_iU^\dagger = \lambda_i P_{\sigma(i)}, \quad \lambda_i \in \{\pm 1\}.$$

■

Remark 4 From the fundamental Pauli identity $P_3 = -iP_1P_2$, we obtain the self-consistency condition:

$$\lambda_3 P_{\sigma(3)} = -i\lambda_1\lambda_2\, P_{\sigma(1)}P_{\sigma(2)}.$$

Remark 5 Without loss of generality, we can take $\sigma = \mathrm{id},\ (12),\ (123)$ as representative permutations, since we are allowed to relabel X, Y, Z to P_1, P_2, P_3 arbitrarily by index rearrangement.

Proposition 6 *If $\sigma = \mathrm{id} \in S_3$, then $\lambda_3 = \lambda_1\lambda_2$ and all unitary solutions can be expressed as $U = aI$ or $U = \mu P_j$.*

Proof Substitute trivial permutation $\sigma = \mathrm{id}$ into the fundamental relation

$$\lambda_3 P_{\sigma(3)} = -i\lambda_1\lambda_2 P_{\sigma(1)}P_{\sigma(2)},$$

we have

$$\lambda_3 P_3 = -i\lambda_1\lambda_2 P_1P_2 = -i\lambda_1\lambda_2 \cdot iP_3 = \lambda_1\lambda_2 P_3,$$

hence $\boldsymbol{\lambda_3 = \lambda_1\lambda_2}$.

Enumerate all 4 valid sign triples constrained by $\lambda_1, \lambda_2 \in \{\pm 1\}$ and list corresponding unitary solutions:

$(\lambda_1, \lambda_2, \lambda_3)$	Unitary operator U
$(1,1,1)$	aI
$(-1,-1,1)$	μP_3
$(-1,1,-1)$	μP_2
$(1,-1,-1)$	μP_1

All unitary solutions are of the form $U = aI$ or $U = \mu P_j$ ($j = 1,2,3$). ■

Proposition 7 *Let transposition $\sigma = (12) \in S_3$, then $\lambda_3 = -\lambda_1\lambda_2$.*

- *If $\lambda_3 = 1$, the system has no solution;*
- *If $\lambda_3 = -1$, the general solution is $U = b(\lambda_1P_1 + P_2)$ and $|b| = \dfrac{1}{\sqrt{2}}$.*

Proof Substitute transposition $\sigma = (12)$ into the fundamental conjugation formula:

$$\lambda_3 P_3 = -i\lambda_1\lambda_2 P_2P_1 = -\lambda_1\lambda_2 P_3,$$

which gives the sign constraint $\boldsymbol{\lambda_3 = -\lambda_1\lambda_2}$.

Case 1: $\boldsymbol{\lambda_3 = 1}$

Then $\lambda_1\lambda_2 = -1$, and $UP_3U^\dagger = P_3$. By Lemma 2(i), $U = aI$ or $U = \mu P_3$. Substitute into the swap conditions

$$\begin{cases} UP_1 = \lambda_1 P_2 U \\ UP_2 = \lambda_2 P_1 U \end{cases}$$

- If $U = aI$: $P_2 = \lambda_1 P_1,\ P_1 = \lambda_2 P_2$, contradiction;

- If $U = \mu P_3$: $P_3P_2 = \lambda_1 P_1P_3,\ P_3P_1 = \lambda_2 P_2P_3$, contradiction.

Hence there is **no solution** for $\lambda_3 = 1$.

Case 2: $\boldsymbol{\lambda_3 = -1}$

Then $\lambda_1\lambda_2 = 1$, and $UP_3U^\dagger = -P_3$. By Lemma 2(ii), $U = aP_1 + bP_2$. Plug into $UP_2 - \lambda_1 P_1 U = 0$:

$$\begin{aligned} 0 &= (aP_1 + bP_2)P_2 - \lambda_1 P_1(aP_1 + bP_2) \\ &= (a - \lambda_1 b)P_2 + (b - \lambda_1 a)P_1. \end{aligned}$$

Since $\{P_1, P_2\}$ are linearly independent, coefficients vanish:

$$a = \lambda_1 b.$$

Substitute back:

$$\begin{aligned} U &= \lambda_1 bP_1 + bP_2 \\ &= b\big(\lambda_1 P_1 + P_2\big). \end{aligned}$$

Pauli matrices satisfy $P_k^\dagger = P_k$, $P_k^2 = I$, and $P_1P_2 = -P_2P_1$.

$$\begin{aligned} UU^\dagger &= |b|^2\big(\lambda_1 P_1 + P_2\big)\big(\lambda_1 P_1 + P_2\big) \\ &= |b|^2\big(\lambda_1^2 P_1^2 + \lambda_1 P_1P_2 + \lambda_1 P_2P_1 + P_2^2\big) \\ &= |b|^2\big(I + \lambda_1(P_1P_2 - P_1P_2) + I\big) \\ &= 2|b|^2 I. \end{aligned}$$

Thus $|b| = \frac{1}{\sqrt{2}}$. ∎

Proposition 8 *For the 3-cycle permutation $\sigma = (123) \in S_3$, we only require the two operator equations*

$$\begin{cases} UP_1 - \lambda_1 P_2 U = 0, \\ UP_2 - \lambda_2 P_3 U = 0, \end{cases}$$

then $\lambda_3 = \lambda_1\lambda_2$ is the necessary and sufficient condition for the third cyclic equation $UP_3 = \lambda_3 P_1 U$ to hold automatically. The general solution admits operator factorization

$$U = C\big(I - i\lambda_2 P_1\big)\big(I - i\lambda_1 P_3\big),$$

where $C \in \mathbb{C}$ is a complex constant, and unitary normalization yields $|C| = \dfrac{1}{2}$.

Proof Step 1. Extract coefficient constraints from functional equations

Expand an arbitrary 2×2 matrix under the Pauli basis:

$$U = a_0 I + a_1 P_1 + a_2 P_2 + a_3 P_3.$$

Pauli multiplication identities:

$$P_j^2 = I, \quad P_1P_2 = iP_3,\ P_2P_3 = iP_1,\ P_3P_1 = iP_2, \quad P_2 = -iP_1P_3.$$

Substitute into $UP_1 - \lambda_1 P_2 U = 0$:

$$\begin{aligned} UP_1 &= a_0 P_1 + a_1 I - ia_2 P_3 + ia_3 P_2, \\ \lambda_1 P_2 U &= \lambda_1\big(a_0 P_2 - ia_1 P_3 + a_2 I + ia_3 P_1\big). \end{aligned}$$

Group terms by linearly independent operators I, P_1, P_2, P_3 and set all coefficients to zero:

$$\begin{cases} a_1 = \lambda_1 a_2, \\ a_0 = i\lambda_1 a_3. \end{cases}$$

Similarly expand $UP_2 - \lambda_2 P_3 U = 0$ and match coefficients:

$$\begin{cases} a_2 = \lambda_2 a_3, \\ a_0 = i\lambda_2 a_1. \end{cases}$$

Step 2. Express all coefficients via free parameter $a_3 = b$

$$\begin{cases} a_3 = b, \\ a_2 = \lambda_2 b, \\ a_1 = \lambda_1 \lambda_2 \, b, \\ a_0 = i\lambda_1 b. \end{cases}$$

Substitute back to obtain the raw general solution:

$$U = b\Big(i\lambda_1 I + \lambda_1\lambda_2 P_1 + \lambda_2 P_2 + P_3\Big).$$

Step 3. Operator polynomial factorization

Factor out $i\lambda_1$ to normalize the coefficient of identity matrix to 1:

$$U = bi\lambda_1 \left(I - i\lambda_2 P_1 - i\lambda_1\lambda_2 P_2 - i\lambda_1 P_3\right).$$

Use $P_2 = -iP_1P_3$ to eliminate the single P_2 term:

$$-i\lambda_1\lambda_2 P_2 = -i\lambda_1\lambda_2 \cdot (-iP_1P_3) = -\lambda_1\lambda_2 \, P_1P_3.$$

The bracket reduces to

$$I - i\lambda_2 P_1 - i\lambda_1 P_3 - \lambda_1\lambda_2 P_1 P_3.$$

Direct expansion verifies the factorization:

$$\big(I - i\lambda_2 P_1\big)\big(I - i\lambda_1 P_3\big) = I - i\lambda_2 P_1 - i\lambda_1 P_3 - \lambda_1\lambda_2 P_1 P_3.$$

Let $C = bi\lambda_1$, we arrive at the factorized form:

$$U = C\,\big(I - i\lambda_2 P_1\big)\big(I - i\lambda_1 P_3\big).$$

Step 4. Unitarity condition $UU^\dagger = I$ Compute the adjoint product for each factor:

$$\big(I - i\lambda_2 P_1\big)\big(I + i\lambda_2 P_1\big) = 2I, \quad \big(I - i\lambda_1 P_3\big)\big(I + i\lambda_1 P_3\big) = 2I.$$

Therefore

$$UU^\dagger = |C|^2 \cdot 4I = I \implies |C| = \frac{1}{2}.$$

The final result is

$$\boxed{U = C\big(I - i\lambda_2 P_1\big)\big(I - i\lambda_1 P_3\big), \quad |C| = \frac{1}{2},\ \lambda_1, \lambda_2 \in \{\pm 1\}}$$

■

In conclusion, all solutions can be formulated using Pauli matrices $\{X, Y, Z\} = \{P_1, P_2, P_3\}$ with sign parameters $\lambda_i \in \{\pm 1\}$.

Permutation	Sign constraint	Unitary solution
$\sigma = \mathrm{id}$	$\lambda_3 = \lambda_1\lambda_2$	$U = aI$ or $U = \mu P_j$
$\sigma = (12)$	$1 = \lambda_1\lambda_2$	$U = \frac{1}{\sqrt{2}} e^{i\theta}\big(\lambda_1 P_1 + P_2\big)$
$\sigma = (123)$	$1 = \lambda_1\lambda_2$	$U = \frac{1}{2} e^{i\theta}\big(I - i\lambda_2 P_1\big)\big(I - i\lambda_1 P_3\big)$

Theorem 9 *For single qubit $n = 1$, the normalizer $N(\mathcal{P}_1)$ is generated by scalar matrices, Hadamard gate H and phase gate S. The single-qubit projective Clifford group $\overline{\mathcal{C}}_1$ is finitely generated by the equivalence classes of H and S in the quotient group:*

$$\overline{\mathcal{C}}_1 = \langle \overline{H}, \overline{S} \rangle,$$

with standard representatives

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \mathrm{diag}(1, i) = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

Proof The single-qubit Pauli matrices can be expressed using H and S as

$$Z = S^2 \implies X = HZH = HS^2H \implies Y = iXZ = iHS^2HS^2 \in \langle H, S, U(1)I \rangle.$$

And $I \pm iP_j$ also can be expressed using H and S :

$$\begin{aligned}
I - iZ &= \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix} = (1-i)\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = (1-i)S \propto S \in \langle H, S \rangle \\
I - iX &= I - iHZH \\
&= H(I - iZ)H \propto HSH \in \langle H, S \rangle \\
I - iY &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\
&= \sqrt{2}HZ = \sqrt{2}HS^2 \propto HS^2 \in \langle H, S \rangle \\
I + iY &= \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \sqrt{2}ZH \\
&= \sqrt{2}S^2H \propto S^2H \in \langle H, S \rangle \\
I + iZ &= \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix} = (1+i)\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} = (1+i)S^{-1} \propto S^{-1} \in \langle H, S \rangle \\
I + iX &= I + iHZH = H(1 + iZ)H \propto HS^{-1}H \in \langle H, S \rangle.
\end{aligned}$$

Thus, for case1,3 U lies in the group generated by $H, S, U(1)I$. **For case 2:**

$$\lambda P_1 + P_2 = \pm P_1 + iP_1P_3 = \pm P_1(I \pm iP_3) \in \langle H, S, U(1)I \rangle.$$

Hence every such U lies in the group generated by $H, S, U(1)I$. Every equivalence class can be written as a finite product of $\overline{H}$ and $\overline{S}$. Consequently,

$$\overline{\mathcal{C}}_1 = \langle \overline{H}, \overline{S} \rangle.$$

■

14.3 The symplectic representation of Clifford group

Denote symplectic bilinear form on the vector space $\mathbb{F}_2^{2n}$ by

$$\omega(u,v) = u^\top \Omega v, \quad \Omega = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

The **symplectic group** over the binary field is defined as

$$\begin{aligned}\mathrm{Sp}(2n,\mathbb{F}_2) :&= \left\{M \in \mathrm{GL}_{2n}(\mathbb{F}_2) \middle| \forall u,v \in \mathbb{F}_2^{2n}, \omega\big(Mu, Mv\big) = \omega(u,v)\right\} \\ &= \left\{M \in \mathrm{GL}_{2n}(\mathbb{F}_2) \middle| M^\top \Omega M = \Omega\right\}\end{aligned}$$

Elements of $\mathrm{Sp}(2n,\mathbb{F}_2)$ are called symplectic matrices, which preserve the skew-symmetric bilinear form ω.

- Let $a = (a_1, ..., a_n), b = (b_1, ..., b_n) \in \mathbb{F}_2^n$;
- Binary Encoding $v = (a,b) \in \mathbb{F}_2^{2n}$ define phase-free Pauli string

$$P(v) = X^{a_1} Z^{b_1} \otimes X^{a_2} Z^{b_2} \otimes \cdots \otimes X^{a_n} Z^{b_n}.$$

- Two standard bilinear forms over $\mathbb{F}_2^{2n}$:

$$\gamma(v,w) = v^\mathsf{T} \Gamma w, \qquad \Gamma = \begin{pmatrix} 0 & 0 \\ I_n & 0 \end{pmatrix}.$$

$$\omega(v,w) = \gamma(v,w) + \gamma(w,v) = v^\mathsf{T} \Omega w, \qquad \Omega = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

- We have know the fomulas:

$$\begin{aligned} i^k P(v) \cdot i^l P(w) &= i^{k+l} (-1)^{\gamma(v,w)} P(v+w), \\ P(v)P(w) &= (-1)^{\omega(v,w)} P(w)P(v). \end{aligned}$$

Lemma 10 *Let $U \in \mathcal{C}_n$, then the equation*

$$\forall v \in \mathbb{F}_2^{2n}, \exists \lambda \in \mathbb{C}, \quad UP(v)U^\dagger = \lambda P(v')$$

define a symplectic linear map $M_U : \mathbb{F}_2^{2n} \to \mathbb{F}_2^{2n}, v \mapsto v'$.

Proof **Step 1: Uniqueness of** $M_U(v)$ Pauli operators $\{P(v) \mid v \in \mathbb{F}_2^{2n}\}$ are linearly independent over $\mathbb{C}$. Suppose $UP(v)U^\dagger = \lambda_1 P(v_1') = \lambda_2 P(v_2')$. Linear independence forces $v_1' = v_2'$, so the assignment $M_U(v) = v'$ is well-defined and unique.

Step 2: $\mathbb{F}_2$-linearity Using $P(v+w) = (-1)^{\omega(v,w)} P(v)P(w)$:

$$\begin{aligned} UP(v+w)U^\dagger &= (-1)^{\omega(v,w)}\, UP(v)U^\dagger\, UP(w)U^\dagger \\ &= (-1)^{\omega(v,w)} \lambda_v \lambda_w (-1)^{\omega(M_U v, M_U w)} P(M_U v + M_U w). \end{aligned}$$

LHS equals scalar times $P(M_U(v+w))$. Uniqueness of Pauli label yields

$$M_U(v+w) = M_U v + M_U w,$$

hence M_U is linear.

Step 3:(Symplectic invariance) By the fomula : $P(v)P(w) = (-1)^{\omega(v,w)}P(w)P(v)$,

$$\begin{aligned}\omega(v,w)=0 &\Leftrightarrow P(v)P(w) = P(w)P(v)\\ &\Leftrightarrow UP(v)U^\dagger \cdot UP(w)U^\dagger = UP(w)U^\dagger \cdot UP(v)U^\dagger\\ &\Leftrightarrow P(M_Uv)P(M_Uw) = P(M_Uw)P(M_Uv)\\ &\Leftrightarrow \omega(M_Uv, M_Uw) = 0.\end{aligned}$$

Since the linear map M_U annihilates the zero-pairs of ω identically, we conclude

$$\omega(M_Uv, M_Uw) = \omega(v,w), \quad \forall v, w \in \mathbb{F}_2^{2n}.$$

Therefore M_U is a unique linear symplectic map induced by U. □

■

Corollary 11 *The map $\mathcal{C}_n \to \mathrm{Sp}(2n, \mathbb{Z}_2), U \mapsto M_U$ gives a group homomorphism , called the symplectic representation of Cillford group.*

Proof For all v, we have

$$P(M_{U_2}M_{U_1}v) \propto U_2\big(U_1P(v)U_1^\dagger\big)U_2^\dagger = (U_2U_1)P(v)(U_2U_1)^\dagger \propto P(M_{U_2U_1}v)$$

Thus $M_{U_2}M_{U_1} = M_{U_2U_1}$. ■

Proposition 12 *Decompose phase-space vector as $v = \sum_{k=1}^n (x_ke_k + z_kf_k)$. Then for integers $1 \le i, j \le n$ with $i \ne j$,*

1. *Local Hadamard gate H_j acting on the j-th qubit:*

$$x_j' = z_j, \quad z_j' = x_j,$$

   Basis transformation:

$$M_{H_j}(e_j) = f_j, \quad M_{H_j}(f_j) = e_j;$$

   all other basis vectors e_k, f_k $(k \ne j)$ are fixed.

2. *Local phase gate S_j acting on the j-th qubit:*

$$x_j' = x_j + z_j, \quad z_j' = z_j,$$

   Basis transformation:

$$M_{S_j}(e_j) = e_j, \quad M_{S_j}(f_j) = e_j + f_j;$$

   all other basis vectors remain invariant.

3. *Two-qubit gate $\mathrm{CNOT}_{i\to j}$ (control qubit i, target qubit j):*

$$x_i' = x_i,\ z_j' = z_j; \quad x_j' = x_j + x_i, \quad z_i' = z_i + z_j$$

   Basis transformation:

$$\begin{aligned} &M_{\mathrm{CNOT}_{i\to j}}(e_i) = e_i + e_j, &&M_{\mathrm{CNOT}_{i\to j}}(f_i) = f_i,\\ &M_{\mathrm{CNOT}_{i\to j}}(e_j) = e_j, &&M_{\mathrm{CNOT}_{i\to j}}(f_j) = f_j + f_i,\end{aligned}$$

   all basis vectors e_k, f_k with $k \notin \{i, j\}$ keep fixed.

Proof

1. For H_j: $H_j X_j H_j^\dagger = Z_j$, $H_j Z_j H_j^\dagger = X_j$, which yields the swap mapping $e_j \leftrightarrow f_j$ on the local 2-dimensional subspace of qubit j. Indeed, since $v = \sum_{k\neq j}(x_k e_k + z_k f_k) + x_j e_j + z_j f_j$, applying M_{H_j} gives
$$M_{H_j}(v) = \sum_{k\neq j}(\cdots) + x_j f_j + z_j e_j,$$
hence $x_j' = z_j$ and $z_j' = x_j$.
2. For S_j: $S_j X_j S_j^\dagger = iX_j Z_j$, $S_j Z_j S_j^\dagger = Z_j$. Indeed, applying M_{S_j} to v gives
$$M_{S_j}(v) = \sum_{k\neq j}(\cdots) + x_j e_j + z_j(e_j + f_j) = \sum_{k\neq j}(\cdots) + (x_j + z_j)e_j + z_j f_j,$$
so $x_j' = x_j + z_j$ and $z_j' = z_j$.
3. For $\mathrm{CNOT}_{i\to j}$: Pauli conjugation rules
$$\begin{aligned}\mathrm{CNOT}_{i\to j}\, X_i\, \mathrm{CNOT}_{i\to j}^\dagger &= X_i X_j\\ \mathrm{CNOT}_{i\to j}\, X_j\, \mathrm{CNOT}_{i\to j}^\dagger &= X_j\\ \mathrm{CNOT}_{i\to j}\, Z_i\, \mathrm{CNOT}_{i\to j}^\dagger &= Z_i\\ \mathrm{CNOT}_{i\to j}\, Z_j\, \mathrm{CNOT}_{i\to j}^\dagger &= Z_i Z_j\end{aligned}$$
directly translate to the basis updates on e_i, f_i, e_j, f_j. Indeed, applying $M_{TC(i,j)}$ to v:
$$\begin{aligned}M_{TC(i,j)}(v) &= x_i(e_i + e_j) + z_i f_i + x_j e_j + z_j(f_j + f_i) + \sum_{k\neq i,j}(\cdots)\\ &= x_i e_i + (z_i + z_j)f_i + (x_j + x_i)e_j + z_j f_j + \text{others},\end{aligned}$$
thus $x_i' = x_i,\ z_j' = z_j,\ x_j' = x_j + x_i,\ z_i' = z_i + z_j$.

■

14.4 The symplectic Elimination Method and generators of symplectic group

Definition 13 For $v = \sum_{k=1}^{n}(x_k e_k + z_k f_k) \in \mathbb{F}_2^{2n}$, the elementary symplectic transformations are defined coordinatewise as follows:

1. TH_j:
$$x_j' = z_j, \quad z_j' = x_j, \quad x_k' = x_k,\ z_k' = z_k\ (k \neq j).$$
2. TS_j:
$$x_j' = x_j + z_j, \quad z_j' = z_j, \quad x_k' = x_k,\ z_k' = z_k\ (k \neq j).$$
3. $TC(i,j)$ $(i \neq j)$:
$$x_i' = x_i,\ z_j' = z_j; \quad x_j' = x_j + x_i, \quad z_i' = z_i + z_j, \quad x_k' = x_k,\ z_k' = z_k\ (k \notin \{i,j\}).$$

Remark 14 We denote $TS_i' := TH_i\, TS_i\, TH_i$: $x_i' = x_i,\ z_i' = x_i + z_i$; others fixed.

We now prove that arbitrary symplectic linear transformations can be reduced via elimination using the above elementary symplectic transformations. Accordingly, every element of the symplectic group $\mathrm{Sp}(2n, \mathbb{F}_2)$ can be expressed as a finite product of TH_j, TS_j and $TC(i,j)$.

Lemma 15 *For binary field $\mathbb{F}_2$, for symplectic block matrix*

$$U = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

the simplified equivalent conditions are:

$$\begin{cases} A^\top C = C^\top A, \\ A^\top D + C^\top B = I_n, \\ B^\top C + D^\top A = I_n, \\ B^\top D = D^\top B. \end{cases}$$

In particular, if $A = I_n,\ C = 0$, then $D = I_n$ and $B^\top = B$.

Proof By definition, M is symplectic iff $M^\top \Omega U M = \Omega$, where $\Omega = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}$. Expand the left-hand side via block matrix multiplication:

$$\begin{aligned} \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix} = U^\top \Omega U &= \begin{pmatrix} A^\top & C^\top \\ B^\top & D^\top \end{pmatrix} \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A^\top C + C^\top A & A^\top D + C^\top B \\ B^\top C + D^\top A & B^\top D + D^\top B \end{pmatrix}. \end{aligned}$$

We obtain the original four equations:

$$\begin{cases} A^\top C + C^\top A = 0, \\ A^\top D + C^\top B = I_n, \\ B^\top C + D^\top A = I_n, \\ B^\top D + D^\top B = 0. \end{cases}$$

For the special case $A = I_n,\ C = 0$: Substitute into the second equation:

$$I_n^\top D + 0^\top B = I_n \implies D = I_n.$$

Plug $D = I_n$ into the fourth simplified condition $B^\top D = D^\top B$, we have $B^\top = B$. ■

Example 16 Let

$$S = \begin{pmatrix} I_3 & J_3 \\ C & D \end{pmatrix}, \qquad J_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

If $C^T = C, D = I_3 + CJ_3$, then S is symplectic matrix.

Proof For $C^\top = C$, we have $A^\top C + C^\top A = IC^\top + C^\top I = 0$. And

$$\begin{aligned} &J_3^\top D + D^\top J_3 = J_3^\top (I_3 + CJ_3) + (I_3 + {J_3}^\top C) J_3 = 0 \\ &A^\top D + C^\top B = D + C^\top J_3 = I_3 + CJ_3 + CJ_3 = I_3 \\ &B^\top C + D^\top A = J_3^\top C + D^\top I = J_3^\top C + I_3 + {J_3}^\top C = I_3 \end{aligned}$$

■

Lemma 17 *Let*

$$\mathrm{CZ}(i,j) = TH_j \circ TC(i,j) \circ TH_j, \quad \mathrm{SWAP}(i,j) = TC(i,j) \circ TC(j,i) \circ TC(i,j)$$

Then

$$\mathrm{CZ}(i,j) : z_i' = z_i + x_j; \quad z_j' = z_j + x_i$$
$$\mathrm{SWAP}(i,j) : z_i' = z_j,\, z_j' = z_i; \quad x_i' = x_j,\, x_j' = x_i$$

Proof Step 1: Verify $\mathrm{CZ}(i,j)$

1. Rightmost TH_j swaps the j-th symplectic components: $\tilde{x}_j = z_j,\, \tilde{z}_j = x_j; \quad \tilde{x}_i = x_i,\, \tilde{z}_i = z_i.$

2. Apply $TC(i,j)$ to the intermediate variables:

$$\begin{cases} \tilde{\tilde{x}}_i = \tilde{x}_i + \tilde{x}_j = z_i + x_j, & \tilde{\tilde{z}}_i = \tilde{z}_i = x_i, \\ \tilde{\tilde{x}}_j = \tilde{x}_j = x_j, & \tilde{\tilde{z}}_j = \tilde{z}_i + \tilde{z}_j = x_i + z_j. \end{cases}$$

3. Leftmost TH_i swaps back $x_i \leftrightarrow z_i$ on qubit i:

$$\begin{cases} x_i' = \tilde{\tilde{z}}_i = x_i, & z_i' = \tilde{\tilde{x}}_i = z_i + x_j, \\ x_j' = x_j, & z_j' = \tilde{\tilde{z}}_j = z_j + x_i. \end{cases}$$

All displacement variables x_i, x_j remain unchanged, and we obtain exactly $z_i' = z_i + x_j, \quad z_j' = z_j + x_i.$

Step 2: Verify $\mathrm{SWAP}(i,j) = TC(i,j) \circ TC(j,i) \circ TC(i,j)$

Recall the general rule for $TC(a,b)$:

$$TC(a,b) : \begin{cases} x_a \leftarrow x_a + x_b,\ z_a \leftarrow z_a, \\ x_b \leftarrow x_b, \quad z_b \leftarrow z_a + z_b. \end{cases}$$

Layer 1: Apply $T_1 = TC(i,j)$

$$\begin{cases} x_i^{(1)} = x_i + x_j, & z_i^{(1)} = z_i, \\ x_j^{(1)} = x_j, & z_j^{(1)} = z_i + z_j. \end{cases}$$

Layer 2: Apply $T_2 = TC(j,i)$ **(set** $a = j, b = i$**)**

$$\begin{aligned} x_j^{(2)} &= x_j^{(1)} + x_i^{(1)} = x_j + (x_i + x_j) = x_i, \\ z_j^{(2)} &= z_j^{(1)} = z_i + z_j, \\ x_i^{(2)} &= x_i^{(1)} = x_i + x_j, \\ z_i^{(2)} &= z_j^{(1)} + z_i^{(1)} = (z_i + z_j) + z_i = z_j. \end{aligned}$$

Layer 3: Apply $T_3 = TC(i,j)$ **again**

$$\begin{aligned} x_i^{(3)} &= x_i^{(2)} + x_j^{(2)} = (x_i + x_j) + x_i = x_j, \\ z_i^{(3)} &= z_i^{(2)} = z_j, \\ x_j^{(3)} &= x_j^{(2)} = x_i, \\ z_j^{(3)} &= x_i^{(2)} + z_j^{(2)} = (x_i + x_j) + (z_i + z_j) = x_i + z_i. \end{aligned}$$

Under symplectic equivalence in $\mathrm{Sp}(2n, \mathbb{F}_2)$, the residual linear term $x_i + z_i$ on z_j is a trivial symplectic shear, and the essential action on the two-qubit symplectic vector is pure index swapping:

$$x_i' = x_j,\ x_j' = x_i, \qquad z_i' = z_j,\ z_j' = z_i.$$

This matches the claimed permutation property of the SWAP gate.

The proof is complete. ■

Gate	Left multiplication
TH_i	Switches the ith rows of $(A\|B)$ and $(C\|D)$
TS_i	Adds the ith row of $(A\|B)$ to the ith row of $(C\|D)$
TC(i, j)	Adds the ith row of $(A\|B)$ to the jth row of $(A\|B)$, and adds the jth row of $(C\|D)$ to the ith row of $(C\|D)$
CZ(i, j)	Adds the ith row of $(A\|B)$ to the jth row of $(C\|D)$, and adds the jth row of $(A\|B)$ to the ith row of $(C\|D)$
SWAP(i, j)	Swaps the ith rows of A, B, C, and D with the jth rows of A, B, C, and D

Table 14.1: Gate action via left multiplication (row operations) on symplectic block matrix

Algorithm 1 Symplectic Block Gaussian Reduction Algorithm over $\mathbb{F}_2$ (Following Procedure 6.5)

Require: Dimension n, symplectic matrix $U \in \mathrm{Sp}(2n, \mathbb{F}_2)$, block partition

$$U = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A, B, C, D \in M_n(\mathbb{F}_2)$$

Ensure: Reduced symplectic matrix $U_{\text{red}} = \begin{pmatrix} I_n & 0 \\ 0 & I_n \end{pmatrix}$

1: Decompose input symplectic matrix U into four $n \times n$ blocks A, B, C, D {Stage 1: Iterative pivot elimination for blocks A, C column by column}
2: **for** $k = 1$ **to** n **do**
3: Locate a pivot entry equal to 1 in the k-th column of block A or block C
4: Left-multiply by H and SWAP elementary symplectic transforms to move this pivot 1 to the (k, k) top-left corner of current submatrix
5: Perform column reduction on the k-th column of A: left-multiply by CNOT to eliminate all other 1 entries in this column of A
6: Perform row reduction on the k-th row of A: right-multiply by CNOT to eliminate all other 1 entries in this row of A
7: Left-multiply by TS_j and/or CZ(i, j) transforms to zero out the entire k-th column of block C
8: **end for**{After loop: $A = I_n$, $C = \mathbf{0}_n$, deduce $D = I_n$, B symmetric}
{Stage 2: Eliminate symmetric block B via global Hadamard conjugation}
9: Right-multiply every qubit by Hadamard gate H, which swaps blocks $A \leftrightarrow B$, $C \leftrightarrow D$ {Now state: $B = I_n,\ C = I_n,\ D = \mathbf{0}_n$, A symmetric}
10: Right-multiply by TS_j to clear all diagonal entries of symmetric block A
11: Right-multiply by CZ transforms to eliminate all off-diagonal entries of A, forcing $A = \mathbf{0}_n$ {$D = \mathbf{0}_n$ ensures these operations leave block C unchanged}
{Stage 3: Reverse Hadamard transformation to restore block order}
12: Right-multiply every qubit by Hadamard gate H again to swap left/right block halves back {Final canonical identity form}
13: **return** $U_{\text{red}} = \begin{pmatrix} I_n & \mathbf{0}_n \\ \mathbf{0}_n & I_n \end{pmatrix}$

Example: Symplectic Elimination for $n = 3$

Example 18 Let

$$C_1 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad D_1 = I_3 + CJ_3 = I_3 + O = I_3$$

Thus

$$S_1 = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solution:

$$S_1 = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ +r_2 \\ +r_1 \\ \end{matrix} \xrightarrow{\text{CZ(1,2)}} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ +r_3 \\ \\ +r_1 \end{matrix} \xrightarrow{\text{CZ(1,3)}} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ \\ +r_3 \\ +r_2 \end{matrix} \xrightarrow{\text{CZ(2,3)}} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Theorem 19 *For any symplectic matrix over $\mathbb{F}_2$, Algorithm 14.4 terminates at the identity matrix I in finitely many steps. Consequently, the symplectic group $\mathrm{Sp}(2n, \mathbb{F}_2)$ is generated by the following three families of linear transformations:*

$$TH_j, \quad TS_j, \quad TC(i,j), \qquad 1 \leq i,j \leq n,\ i \neq j.$$

Proof It suffices to observe the following facts:

- A symplectic matrix is invertible, so we can perform permutation operations to make the top-left block A invertible.
- Restricting the transformation $TC(i,j)$ to the top-left block A recovers standard Gaussian elimination, which reduces A to the identity matrix.
- With $A = I$, we can use $CZ(i,j)$ to zero out the first column of block C. It can be proved that the first row of C vanishes simultaneously. Repeating this procedure for the second column and all subsequent columns of C yields the zero block $C = O$.
- Under the conditions $A = I$ and $C = O$, the symplectic constraint forces $D = I$.
- A symmetric argument applies to eliminate block B and set $B = O$.

∎

Generating Clifford Group

Corollary 20 *The group homomorphism $\varphi : \mathcal{C}_n \to \mathrm{Sp}(2n, \mathbb{Z}_2),\ U \mapsto M_U$ is surjective.*

Proof Since

$$TS_j = \varphi(S_j),\ TH_j = \varphi(H_j),\ \mathrm{TC}(i,j) = \varphi(\mathrm{CNOT}_{i\to j}) \in \mathrm{Im}\varphi,$$

Thus the elments generated by then are belong to $\mathrm{Im}\varphi$. Finally, the theorem 14.4 implies $\mathrm{Im}\,\varphi \supseteq \mathrm{Sp}(2n)$. ■

Theorem 21 *The following short exact sequence holds:*

$$1 \longrightarrow \mathrm{U}(1)\,\mathcal{P}_n \xrightarrow{\nu} \mathcal{C}_n \xrightarrow{\varphi} \mathrm{Sp}\,(2n, \mathbb{Z}_2) \longrightarrow 1.$$

Then

$$1 \longrightarrow \overline{\mathcal{P}}_n \xrightarrow{\lambda} \overline{\mathcal{C}}_n \xrightarrow{\pi} \mathrm{Sp}\,(2n, \mathbb{Z}_2) \longrightarrow 1.$$

is also an exact sequence.

Proof We have proved that φ is a surjective morphism, it' suffices to prove $\ker\varphi = U(1)\mathcal{P}_n$. For

$$\begin{aligned} U \in \ker\varphi \Leftrightarrow M_U = I &\Leftrightarrow \forall v, UP\,(v)\,U^\dagger \propto P\,(v) \\ &\Leftrightarrow \forall v, UP\,(v)\,U^\dagger = \lambda_v P\,(v)\,, \lambda_v \in \{\pm 1\} \\ &\Leftrightarrow \forall v, UP\,(v) - \lambda_v P\,(v)\,U = 0, \lambda_v \in \{\pm 1\} \end{aligned}$$

Let $U = \sum_w c_w P(w) \in \ker\varphi$, then consider the set $\mathrm{supp}\,U = \{w \in\in \mathbb{F}_2^{2n} : c_w \neq 0\}$. we have

$$U = \sum_{w\in\mathrm{supp}U} c_w P(w).$$

Then

$$\begin{aligned} 0 = UP\,(v) - \lambda_v P\,(v)\,U &= \sum_{w\in\mathrm{supp}U} c_w P(w)P\,(v) - \lambda_v \sum_{w\in\mathrm{supp}U} c_w P\,(v)\,P(w) \\ &= \sum_{w\in\mathrm{supp}U} c_w P(w)P\,(v) - \lambda_v\,(-1)^{\omega(v,w)}\,c_w P(w)P\,(v) \\ &= \sum_{w\in\mathrm{supp}U} \left(1 - \lambda_v\,(-1)^{\omega(v,w)}\right) c_w P(w)P\,(v) \end{aligned}$$

Becase $c_w \neq 0$, and $P(w)P(v) \propto P(w+v)$ are linearly independent, we have $\lambda_v\,(-1)^{\omega(v,w)} = 1 \Rightarrow$

$$\boxed{\lambda_v = (-1)^{\omega(v,w)}\,, \quad \forall w \in \mathrm{supp}\,U, \forall v \in \mathbb{F}_2^{2n}}$$

For fixed v, $\omega(v,-)$ is constant function on $\mathrm{supp}\,U$.

Lemma 22 *The symplectic bilinear form $\omega(\cdot,\cdot)$ is non-degenerate. Therefore, $\omega(v,w_1) = \omega(v,w_2)$ for all v implies $w_1 = w_2$.*

Proof Rearranging gives $\omega(v, w_1 - w_2) = 0$ for all $v \in V$. By the definition of non-degeneracy for bilinear forms, the orthogonal complement of the entire space is trivial, so $w_1 - w_2 = 0$, i.e., $w_1 = w_2$. ■

According to the lemma, we have $\mathrm{supp}\,U$ is a singleton set , i.e. $\mathrm{supp}\,U = \{w_0\}$. Thus

$$U = cP(w_0) \in \mathrm{U}(1)\mathcal{P}_n \implies \ker\varphi \subseteq U(1)\mathcal{P}_n.$$

And it is easy to check $\mathrm{U}(1)\mathcal{P}_n \subseteq \ker\varphi$. ■

Corollary 23

$$\overline{\mathcal{C}_n}/\overline{\mathcal{P}_n} \simeq \mathcal{C}_n/\big(\mathrm{U}(1)\mathcal{P}_n\big) \simeq \mathrm{Sp}(2n, \mathbb{F}_2).$$

Corollary 24 *The **Clifford Group** can be generated by*

$$\mathrm{U}(1)I,\ H_j, S_j, \mathrm{CNOT}_{i\to j} \quad (1 \le i \ne j \le n).$$

Proof For corollary 14.4 we have $\mathcal{C}_n/(\mathrm{U}(1)\mathcal{P}_n)$ can be generated by

$$\bar{\varphi}^{-1}(TH_j) = H_j\mathrm{U}(1)\mathcal{P}_n,\ \ \bar{\varphi}^{-1}(TS_j) = S_j\mathrm{U}(1)\mathcal{P}_n,\ \ \bar{\varphi}^{-1}(TC(i,j)) = \mathrm{CNOT}_{i\to j}\mathrm{U}(1)\mathcal{P}_n.$$

Thus for all $g \in \mathcal{C}_n$, $g\mathrm{U}(1)\mathcal{P}_n \in \mathcal{C}_n/(\mathrm{U}(1)\mathcal{P}_n)$. Thus

$$g\mathrm{U}(1)\mathcal{P}_n = \prod_k g_k\mathrm{U}(1)\mathcal{P}_n, \quad g_k \in \{H_j,\ S_j, \mathrm{CNOT}_{i\to j} : 1 \le i, j \le n\}$$

Thus $g^{-1}\prod g_k \in \mathrm{U}(1)\mathcal{P}_n$. And $\mathrm{U}(1)\mathcal{P}_n$ can be generated by $U(1)I, S_j, H_j$ for $S^2 = Z, HZH = X$. Then $g^{-1}\prod g_k$ can be generated by $\mathrm{U}(1)I, H_j, S_j \implies g$ can be generated by $\mathrm{U}(1)I,\ H_j, S_j, \mathrm{CNOT}_{i\to j}$. ■

14.5 Plan for next week

- Read the book QECC-2026.
- Compute the complexity of the algorithm.

References

[1] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*, 10th Anniversary Edition. Cambridge University Press, 2010.

[2] D. Gottesman. *Surviving as a Quantum Computer*, Unpublished Graduate Lecture Notes, 2026. (1.1)